Adnane Bejja

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Education

Concordia University **Expected May 2027**
Bachelor of Computer Science *Montreal, QC*

Work Experience

Intact Financial Corporation **January 2025 – Present**
Software Developer I *Montreal, QC*

- Engineered, optimized, and maintained **Java/Spring Boot** microservices and **REST APIs** in an **Agile SDLC**, writing **clean code** supporting **500K+ monthly transactions** and improving system **efficiency and scalability** by reducing average response times by **44%** through API and database query optimization.
- Designed the **software architecture** and implemented **multi-session support** using **JavaScript**, **sessionStorage**, and unique tab identifiers to isolate application state across browser tabs, applying **system design** principles to enable concurrent policy workflows and reduce transaction handling time by **35%**.
- Enhanced production **observability** with **MDC-based contextual logging** and troubleshot high-impact production systems using **Dynatrace**, **Kibana**, and **Centralized Log Management (CLM)**, reducing a recurring application-crashing error from **5,000+ to approximately 160 occurrences per month**.
- Built and integrated **Kafka**-based **real-time event-driven data processing workflows** within a production **distributed system**, applying **concurrent and parallel processing** techniques to handle thousands of asynchronous messages while maintaining fault isolation, scalability, stability, and system reliability.
- Developed and maintained **automated unit and component tests** achieving **90% coverage**, resolved **30+ production defects**, eliminated all critical **SonarQube** findings, and participated in **Git-based code reviews**, incorporating engineering feedback and established software development best practices.

Projects

SoccerVision | *Source Code* **Python, PyTorch, OpenCV, YOLO, ByteTrack, CUDA**

- Built an end-to-end **computer vision pipeline** processing **750 frames of 1080p soccer footage**, detecting **4 object classes** and exporting **17,904 tracking records** with team assignments and spatial analytics.
- Accelerated **PyTorch inference** with **NVIDIA CUDA**, increasing throughput from **4.35 to 53.15 FPS** (**12.2**×) using the same YOLO model and identical benchmark frames.
- Evaluated an **object detection model** on **599 annotated objects**, achieving **81.5% overall mAP@50** and **98.2% player mAP@50** on an in-domain test set.

7Eats | *Source Code* | *Live App* **Next.js, React, TypeScript, PostgreSQL**

- Collaborated with a team of **3 developers** to build and deploy a full-stack food marketplace using **Next.js**, **React**, **TypeScript**, **PostgreSQL (Neon)**, and **Drizzle ORM**.
- Integrated **Stripe Connect** for marketplace payment splitting, cook payouts, platform fees, refunds, and transaction snapshotting for reliable financial transaction processing.
- Implemented **role-based authentication and authorization** with **Better Auth** and built location-aware discovery using **Mapbox**, with server-side validation for promotions and discount workflows.

Awards

ConUHacks IX Winner | *DevPost Link* **2025**

- Won **1st place** in the Beenox Challenge at Quebec's largest hackathon (**800+ participants**) by collaborating in a fast-paced team environment and solving technical problems under strict delivery constraints.
- Prototyped and engineered a **real-time multiplayer networking game** in under **24 hours** using raw **Java**, applying **data structures and algorithms**, **object-oriented programming**, and problem-solving under technical ambiguity.

Technical Skills

- **Languages**: Java, Python, JavaScript, C#, TypeScript, SQL, C++, Kotlin
- **Backend & APIs**: Spring Boot, Backend Application Development, API Development, REST APIs, Microservices, Service-Oriented Architecture (SOA), High-Volume APIs, Event-Driven Architecture (Kafka), Real-Time Systems, Async Processing, Threads
- **Data & Databases**: PostgreSQL, MySQL, MongoDB (NoSQL), Redis, SQL Server Fundamentals, Data Processing, Relational Databases, Caffeine Caching
- **Cloud & DevOps**: AWS, Docker, OpenShift, Render, Jenkins, GitHub Actions, Continuous Integration, CI/CD, Deployment Pipelines, Live Deployments, Git, Version Control, Linux Shell Scripting
- **Testing & Quality**: Clean Code, Automated Testing, JUnit 5, Mockito, Unit Testing, Component Testing, Integration Testing, Test Coverage, SonarQube, Code Reviews
- **Computer Science**: Algorithms & Data Structures, Object-Oriented Programming, System Design, Distributed Systems, Concurrent Programming, Parallel Processing, Design Patterns, Agile SDLC
- **Observability**: Systems Monitoring, Dynatrace, Kibana, Centralized Log Management (CLM), MDC Contextual Logging, Production Troubleshooting, System Reliability & Stability